

15 Easy Level Coding Questions on For Loop Statements

Each question includes scenario, description, input format, output format, constraints, sample input, sample output and explanation.

Question 1: Print Numbers from 1 to N

Scenario

A beginner is learning how to count values using a loop. The task is to print all numbers starting from 1 up to a given number N.

Detailed Description

Write a program that reads an integer N and prints numbers from 1 to N in a single line. Use a for loop to generate each number.

Input Format

One integer N.

Output Format

Print numbers from 1 to N separated by spaces.

Constraints

$1 \leq N \leq 1000$

Sample Input

5

Sample Output

1 2 3 4 5

Explanation

The loop starts at 1 and runs until 5. Each value is printed in order.

Question 2: Print Even Numbers up to N

Scenario

A teacher wants to show students all even roll numbers in a class list up to N.

Detailed Description

Write a program to print all even numbers from 2 to N. Use a for loop and increase the value by 2 each time.

Input Format

One integer N.

Output Format

Print all even numbers up to N separated by spaces. If no even number exists, print nothing.

Constraints

$1 \leq N \leq 1000$

Sample Input

10

Sample Output

2 4 6 8 10

Explanation

The even numbers not greater than 10 are 2, 4, 6, 8 and 10.

Question 3: Sum of First N Natural Numbers

Scenario

A cashier wants to calculate the total token value from 1 to N.

Detailed Description

Write a program that reads N and calculates the sum $1 + 2 + 3 + \dots + N$ using a for loop.

Input Format

One integer N.

Output Format

Print the sum of the first N natural numbers.

Constraints

$1 \leq N \leq 10000$

Sample Input

6

Sample Output

21

Explanation

$1 + 2 + 3 + 4 + 5 + 6 = 21$.

Question 4: Multiplication Table

Scenario

A student wants to quickly generate the multiplication table of a number for practice.

Detailed Description

Write a program that reads an integer N and prints its multiplication table from 1 to 10 using a for loop.

Input Format

One integer N.

Output Format

Print 10 lines in the format N x i = result.

Constraints

1 <= N <= 100

Sample Input

3

Sample Output

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

Explanation

The loop runs from 1 to 10 and multiplies 3 with each loop value.

Question 5: Factorial of a Number

Scenario

A math app needs to calculate factorial values for small positive numbers.

Detailed Description

Write a program that reads N and finds N factorial using a for loop. Factorial means multiplying all numbers from 1 to N.

Input Format

One integer N.

Output Format

Print the factorial of N.

Constraints

$0 \leq N \leq 12$

Sample Input

5

Sample Output

120

Explanation

$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$. For 0, factorial is 1.

Question 6: Count Digits in a Number

Scenario

A banking system wants to know how many digits are present in an account number.

Detailed Description

Write a program to count the number of digits in a given non-negative integer. Use a loop to repeatedly remove the last digit.

Input Format

One non-negative integer N.

Output Format

Print the number of digits in N.

Constraints

$0 \leq N \leq 1000000000$

Sample Input

45678

Sample Output

5

Explanation

The number 45678 has five digits.

Question 7: Reverse a Number

Scenario

A security tool checks the reverse form of a numeric PIN.

Detailed Description

Write a program that reads a number and prints its reverse. Use a loop to take the last digit and build the reversed number.

Input Format

One non-negative integer N.

Output Format

Print the reversed number.

Constraints

$0 \leq N \leq 1000000000$

Sample Input

1234

Sample Output

4321

Explanation

Digits are taken from right to left: 4, 3, 2, 1.

Question 8: Sum of Digits

Scenario

A game gives bonus points based on the sum of digits in a score.

Detailed Description

Write a program to find the sum of digits of a given number using a loop.

Input Format

One non-negative integer N.

Output Format

Print the sum of digits.

Constraints

$0 \leq N \leq 1000000000$

Sample Input

248

Sample Output

14

Explanation

$2 + 4 + 8 = 14$.

Question 9: Print Squares from 1 to N

Scenario

A math worksheet generator needs square values for numbers from 1 to N.

Detailed Description

Write a program that reads N and prints the square of every number from 1 to N.

Input Format

One integer N.

Output Format

Print squares from 1^2 to N^2 separated by spaces.

Constraints

$1 \leq N \leq 1000$

Sample Input

5

Sample Output

1 4 9 16 25

Explanation

The program prints $1*1$, $2*2$, $3*3$, $4*4$ and $5*5$.

Question 10: Print Odd Numbers in a Range

Scenario

A class monitor wants to display all odd-numbered seats between two seat numbers.

Detailed Description

Write a program that reads two integers A and B, then prints all odd numbers from A to B inclusive using a for loop.

Input Format

Two integers A and B.

Output Format

Print odd numbers from A to B separated by spaces.

Constraints

$1 \leq A \leq B \leq 1000$

Sample Input

3 12

Sample Output

3 5 7 9 11

Explanation

The odd numbers in the range 3 to 12 are 3, 5, 7, 9 and 11.

Question 11: Number Divisors

Scenario

A student wants to list all exact divisors of a number.

Detailed Description

Write a program that reads N and prints all numbers that divide N exactly. Use a for loop from 1 to N.

Input Format

One integer N.

Output Format

Print all divisors of N separated by spaces.

Constraints

$1 \leq N \leq 1000$

Sample Input

12

Sample Output

1 2 3 4 6 12

Explanation

12 is divisible by 1, 2, 3, 4, 6 and 12 without remainder.

Question 12: Check Prime Number

Scenario

A number game needs to check whether a selected number is prime.

Detailed Description

Write a program that reads N and checks whether it is prime. A prime number has exactly two factors: 1 and itself.

Input Format

One integer N.

Output Format

Print Prime if N is prime, otherwise print Not Prime.

Constraints

$1 \leq N \leq 10000$

Sample Input

7

Sample Output

Prime

Explanation

7 is divisible only by 1 and 7, so it is prime.

Question 13: Simple Star Line Pattern

Scenario

A design program needs to print a line of stars for decoration.

Detailed Description

Write a program that reads N and prints N stars in one line using a for loop.

Input Format

One integer N.

Output Format

Print N star symbols continuously.

Constraints

$1 \leq N \leq 100$

Sample Input

6

Sample Output

Explanation

The loop runs 6 times and prints one star each time.

Question 14: Increasing Number Pattern

Scenario

A beginner pattern exercise requires printing rows of numbers with increasing length.

Detailed Description

Write a program that reads N and prints a pattern where row i contains numbers from 1 to i.

Input Format

One integer N.

Output Format

Print N rows of increasing number pattern.

Constraints

$1 \leq N \leq 50$

Sample Input

4

Sample Output

```
1
1 2
1 2 3
1 2 3 4
```

Explanation

The first row has 1 number, the second row has 2 numbers, and so on.

Question 15: Calculate Power

Scenario

A calculator app needs to calculate base raised to exponent without using built-in power functions.

Detailed Description

Write a program that reads two integers base and exponent, then calculates $\text{base}^{\text{exponent}}$ using a for loop.

Input Format

Two integers base and exponent.

Output Format

Print the calculated power value.

Constraints

$1 \leq \text{base} \leq 20$, $0 \leq \text{exponent} \leq 10$

Sample Input

2 5

Sample Output

32

Explanation

2^5 means $2 \times 2 \times 2 \times 2 \times 2 = 32$.